Instacart Autonomous Delivery

APM Take Home Challenge

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Why Invest in Autonomous Delivery?

Reduce Fulfillment Costs/Order

- Last mile delivery accounts for **53%** of total delivery costs, with each delivery costing **~$10** to execute against an average charge of only **$8.08** — meaning most deliveries lose money
- **Labor** (drivers) makes up **50-60%** of last mile costs, and autonomous robots can cut per-delivery costs by **40-60%**
- Robots operate around the clock without overtime, benefits, or scheduling constraints, compounding **savings at scale**

Increase Delivery Speed & Consistency

- Today's delivery **windows fluctuate** with shopper availability, distance, traffic, and batching – customers often see wide ranges like "2:00–4:00 PM"
- Robots run **fixed routes at consistent speeds**, eliminating driver-side variability
- Result: narrower, **more reliable ETAs** customers can actually plan around

Deliver Novel Customer Experiences

- Robot delivery is a fundamentally **new interaction** – real-time autonomous vehicle tracking, phone-based cargo unlock, precise arrival times
- Creates a memorable, social-media **shareable moment** that brings brand awareness and **organic marketing**
- For younger, tech-forward customers especially, this is a **branding moment**, not just a logistics improvement

Unlock New Advertising Revenue

- Robots are **moving billboards** — branded wraps, digital screens, and sponsored delivery experiences create an entirely new ad surface for Instacart's retail and CPG partners
- **Extends** Instacart's $1.45B ad platform from screens into the physical world – a differentiated pitch to advertisers already spending on the platform
- Further enhances the futuristic **Act 2 vision "Grocery OS"** that retailers partner with Instacart for

Competitive Urgency

- **Walmart** is scaling drone delivery to reach 75% of the U.S. population
- **DoorDash** is piloting its own proprietary "Dot" robot in Phoenix
- And many more. **If Instacart doesn't offer retailers an autonomy pathway, competitors will**

Fulfillment Cost/Order drops 40-55%, turning delivery from a loss center into a margin contributor.

Cost Component	Human Shopper	Store Associate + Robot
In-store Pick + Pack	~ $4 (gig shopper)	~ $0 (associate - on retailer's payroll)
Last-mile Transport	~ $6 (driver, fuel, insurance)	~ $1.5 - $2.5 (robot amortization)
Total Fulfillment	**~ $10**	**~ $1.5 - $2.5**
Avg. fee charged	**~ $8**	**~ $3.5 - $5**
Net per delivery	**~ - $2 Loss**	**~ + $2 - $3.5 Profit**

The Right Robot for the Right Order

	Sidewalk Robots (ie. Serve, Coco)	Road-Legal Pods (ie. Nuro)	Aerial Drones (ie. Wing)
Best for	Small baskets (<10 kg): fill-ins, restaurant orders, convenience items	Medium-to-large baskets (10–125 kg): weekly grocery, Instacart for Business	Ultra-light, ultra-urgent (<5 lbs): pharmacy, single-item top-ups
Geography	Dense urban cores with pedestrian infrastructure (LA, Miami, campuses)	Suburban grids with wide roads (Phoenix, Dallas, Florida suburbs)	Suburban-to-rural with low air traffic and FAA clearance
Speed	4–7 km/h; ~15–30 min within 2-mile radius	Road-legal speeds; ~15–25 min within 5-mile radius	Sub-15 min within FAA-approved radius
Key Tradeoff	Lowest cost/delivery, but limited payload — can't serve core $75+ orders	Higher unit cost and regulatory complexity, but handles 70% of Instacart volume	Fastest and highest novelty, but tiny payload, FAA overhead, weather dependency
Instacart's Role	Partner fleet (leverage Uber/Serve relationship)	Partner fleet or corridor-specific B2B contracts (Nuro)	Monitor and evaluate – not a near-term priority

Pilot Phase 1:
Sidewalk robots for small-basket orders in dense urban markets – lowest risk, fastest learning.

Pilot Phase 2:
Road-legal pods for suburban grocery – the high-impact opportunity.

Drone delivery only makes sense for "last-minute," small-basket orders of ultra-light items that require high speed and convenience. It is most effective in suburban, low-traffic areas with easy access to landing zones (backyards).

→ Currently not a good match for Instacart's big-basket $75+ orders.

How the Experience Evolves

Current Journey – Human Shopper

Customer Places Order → Match to Shopper → Shopper Drives to Store → Shopper Finds Items → Shopper Checks out & Packs → Shopper Drives & Delivers → Customer Rates & Tips Shopper

Pain Points

- **Inconsistent pick quality** – shoppers rotate across stores they don't know (hours, layout, & inventory)
- **Wide delivery windows** driven by shopper availability and traffic
- **High cost** – one person is paid to drive, shop, and deliver (+ tips)
- Coordination friction across multiple variables = **multiple failure modes**

New Journey – Store Associate + Robot

Customer Places Order (Routes Directly to Retail Store) → Associate Picks & Packs Order → Associate Loads Robot → Straight to Customer (with Real-time Track) → Robot Arrives: Tap to Unlock

Improvements

- **Higher pick quality** – associates know their own store's layout, stock, and products
- **Tighter delivery windows** – robots run fixed routes at consistent speeds, no driver variance
- **Lower cost** – picking and delivery are unbundled, each handled by the lowest-cost option (no tips)
- **Scalable** – reduced touchpoints, fleet size drives capacity, not gig worker recruitment; no peak-hour availability crashes

Risks & Mitigations

Retail Store Associate Capacity

Picking Instacart orders could compete with associates' core responsibilities and slow down fulfillment. Retailers may also push back on absorbing picking labor costs that were previously Instacart's responsibility.

- **Mitigation:**
  Start with low-volume stores where associate capacity has slack.
  Instacart can offset the labor burden through revenue-sharing on delivery fees or per-order incentive payments to the retailer.
  Long-term → dedicated "pick associate" roles.

Negative Public Sentiments

Instacart has ~600,000 shoppers. Any narrative positioning robots as a threat creates PR and operational risk. In Chicago, 83% of surveyed residents opposed robot expansion.

- **Mitigation:**
  Frame as role evolution, not replacement.
  Shoppers remain essential for big-basket orders robots can't serve (70% of current volume).
  For robot-eligible orders, messaging leads with "hand-picked by a store associate" – preserving the human connection customers value.

Sidewalk Safety & Accessibility

A Serve robot weighs 220 lbs empty. Studies show a 60 kg robot at 11 km/h can severely injure a child. One recently crashed into a Chicago bus stop.

- **Mitigation:**
  Accessibility-first design.
  Partner with disability advocacy orgs.
  Mandatory yield protocols.
  Limit zones to ADA-compliant sidewalks.
  Publish a public safety commitment before launch.

Regulatory Fragmentation

Toronto and Ottawa enacted bans citing safety for older adults and people with disabilities. 38 states adopted AI-related measures in 2025, but regulation is intensely local. What works in Phoenix may be illegal in Portland.

- **Mitigation:**
  Start where regulation is favorable and has precedents of ADRs/AVs (Arizona, LA, Florida, Texas, Virginia, etc.)
  Build local safety and community engagement track records as a template for expansion.

The "Big Basket" Payload Gap

70% of Instacart orders are $75+ big-basket shops. Some have large individual items like a 24-pack of water bottles. Drones cannot handle the weight, while current sidewalk robots may handle the weight but usually cannot fit in the compartment.

- **Mitigation:**
  Phase 1 explicitly targets small baskets – we don't pretend the tech fits the core use case.
  Phase 2 (Nuro road-legal pods, 50–125+ kg) tackles real volume.

Vandalism, Theft & Edge Cases

Robots have been kicked, spray-painted, and "bullied" in many cities. Cargo theft and edge cases (construction, detours) are also real.

- **Mitigation:**
  360° cameras, tamper-detection locks, geo-fenced zones, remote human operators for edge cases.
  Start in controlled lower-risk environments before expanding.
  This risk largely relies on the ADR-development company we decide to partner with.

The Pilot Program

HOW?

Partner, Not Build:

Amazon Scout and FedEx Roxo both failed after years of internal hardware R&D.
DoorDash is building its own "Dot" – a high-risk, capital-intensive bet.
Instacart's identity is the operating system for grocery, not a robotics manufacturer.
→ Partner on hardware, build the software integration layer – the same playbook behind Instacart Enterprise.
After pilot, it is important to diversify our vendors to scale faster.

Recommended Partners:

Phase 1

Serve Robotics

- Currently deploying 2,000 Gen-3 robots via Uber Eats. Instacart's existing Uber partnership can create a direct on-ramp.
- Gen-3 produced at 1/3 previous cost via Magna manufacturing.
- Level 4 autonomy.

Phase 2

Nuro

- Road-legal pods, 125–300 kg payload.
- The only vehicle company currently available on market that can handle Instacart's core big-basket volume.

Partnership Integration Work Required:

- **Retail store handoff software** (order staging, robot dispatch, associate workflow)
- **Customer-facing** tracking and unlock experience
- **Routing optimization** integrated with existing delivery logic
- **Data pipeline** connecting robot performance to the platform

WHERE?

Phase 1: LOS ANGELES, CALIFORNIA

- **Car-dependent city** means small-basket trips are less convenient → more identified use cases for delivery services.
- **Weather:** Year-round mild conditions eliminate the #1 robot failure mode
- **Geography:** West Hollywood, Santa Monica, Koreatown, Silver Lake – grid streets, maintained sidewalks, high-density residential within 1–2 mi of retail
- **Regulation:** California permits sidewalk robots; LA has no restrictive local ordinances
- **Existing fleet:** Serve already operating at scale in LA via Uber Eats – mapped routes, trained operators in place
- **Retail density:** Strong Instacart relationships with Sprouts, Gelson's, Ralphs, 7-Eleven, Bristol Farms

SCOPE

90	3-5	< 10kg	15-25	500-1k
day pilot	lower-volume/ high-capacity retail partners	small-basket for groceries + business	Serve Gen-3 robots, 8AM - 8PM	robot-fulfilled deliveries

Phase 2 Preview: Suburban full-basket grocery in Phoenix, AZ using Nuro pods for $75+ orders (70% of volume). Arizona's AV-friendly legislation, DoorDash's market validation, and the city's wide suburban grid make it the natural next proving ground.

MVP Features

Step 1: Opt-In at Checkout

- On the checkout screen, when the order qualifies (under payload limit, within service area), a new toggle appears: "Autonomous Delivery" alongside standard options.
- Clear sub-copy: "Hand-picked by a store associate, delivered by autonomous robot to your curb. No tip charged if matched."
- Optional "Learn More" tooltip with pop-up screen.

Step 2: Order Confirmation & Status

- Picking → Packing → Staging for Robot → Robot En Route → Arriving → Arrived.
- Push notification is sent when an autonomous vehicle has been matched/ on its way along with ETA.
- The familiar shopper-picking flow stays intact.

Step 3: Live Tracking

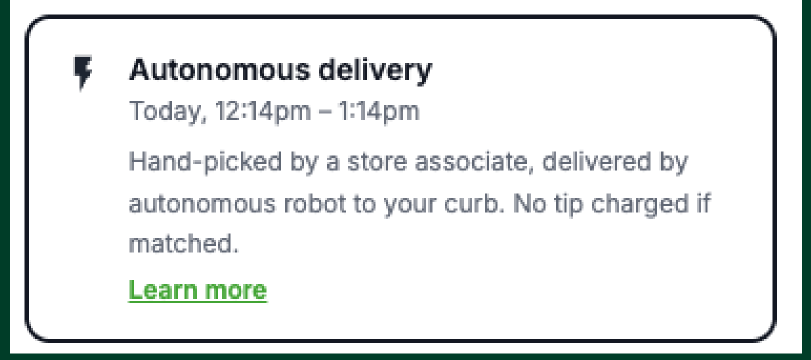
- Map-based tracking
- Robot moves in real time.
- "Help" button for edge cases: "Robot seems stuck?" → remote operator support. "Change drop-off spot?" → limited modification within geofence.

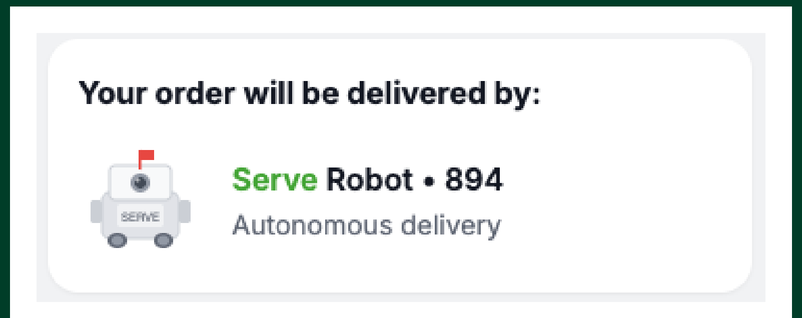
Step 4: Arrival & Unlock

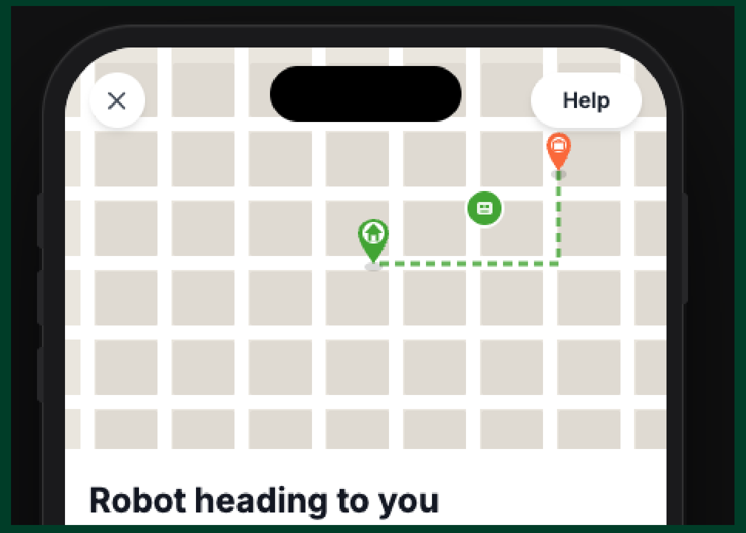
- Push notification: "Your robot has arrived! Head outside to pick up your groceries."
- Prominent "Unlock" button — slide/tap to open cargo bay (Bluetooth/PIN).
- 10-minute pickup timer; reminder if no pickup; robot returns order to store after timeout.

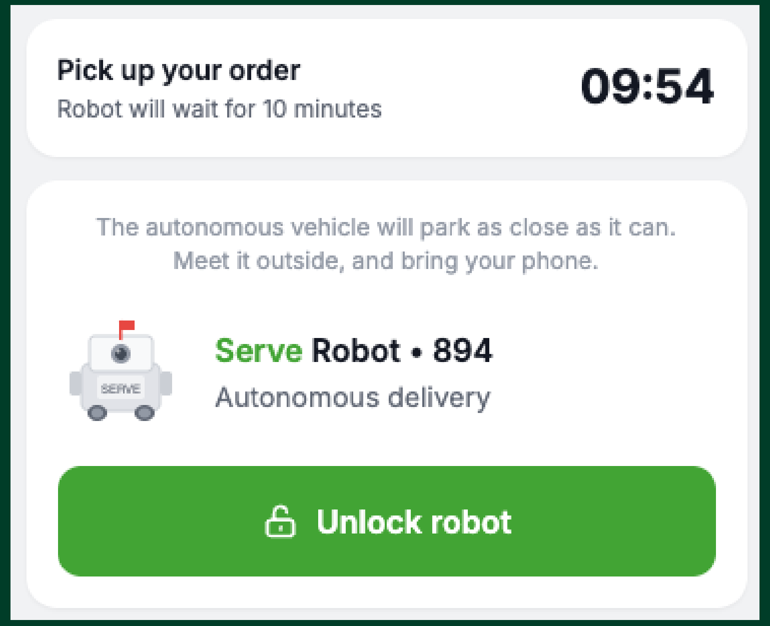
Step 5: Post-Delivery Feedback

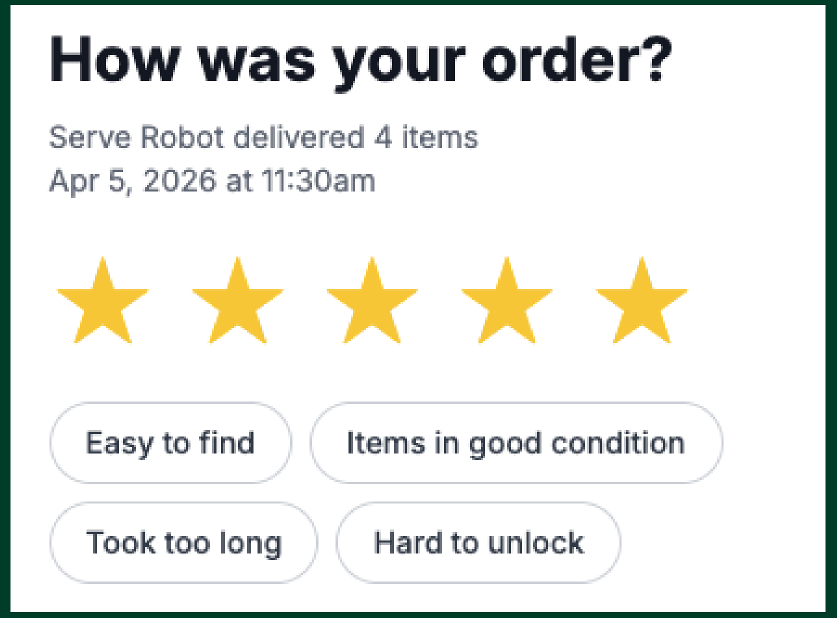
- "How was your robot delivery?" — 1–5 star rating + tags: "Easy to find," "Items in good condition," "Took too long," "Hard to unlock."
- Feeds directly into the pilot learning agenda.











[Link to Prototype](https://lanynguyen.github.io/instacart/dist/)

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Hypotheses & Metrics

Learning Objective	Hypothesis	Key Metric	Target
Will customers choose robot delivery?	≥15% of eligible customers opt in within 60 days	Robot Opt-In Rate (% of eligible orders selecting robot)	15%+; segment by Instacart+, SNAP
Does the experience meet the quality bar?	Robot CSAT within 10% of human-delivered orders by Day 90	Post-Delivery CSAT (1–5 stars) + Delivery Success Rate	≥4.0 stars; ≥95% success rate
Does it actually reduce cost?	40%+ reduction in last-mile transport cost vs. human baseline	Cost Per Delivery (robot amortization + energy + monitoring ÷ deliveries)	<$2.50/delivery; 20+ deliveries/robot/shift
Can store associates absorb the picking workload?	Associates in pilot stores fulfill robot orders without degrading in-store service or exceeding a 10% increase in pick-to-dispatch time vs. gig shoppers	Pick-to-Dispatch Time + Store Operations Score (existing retailer KPIs for shelf stocking, checkout wait, etc.)	Pick-to-dispatch ≤15 min; no decline in store ops metrics
How does the community respond?	Net positive or neutral public sentiment with proactive engagement	Community Sentiment Score (social monitoring + 311 complaints + surveys)	<1 safety incident per 1,000 deliveries

Assumptions:
Serve Gen-3 at $10K/unit, amortized over ~13,000 deliveries (12/day × 3 yr lifespan). Energy $0.18/delivery. Remote monitoring at 1:15 operator ratio = $0.80/delivery at pilot. Maintenance 12%/yr + $750/yr insurance = $0.36/delivery. Total: ~$2.00 at pilot, ~$1.15 at scale.

Go/No-Go Framework:
If Phase 1 validates customer willingness, cost improvement, associate feasibility, and community acceptance → expand to Phase 2 (suburban full-basket). If not → we've learned at low cost where the technology and market aren't ready.